

Week 5 Report

Held-out colour-model selection and multicolour pipeline validation

Progress this week

- Reviewed Ihor's coded-marker scale implementation and PR 19 against the colour-calibration branch. The scale protocol is complementary to PR 18 and has no direct file conflict, but I identified two edge cases that should be resolved before relying on its train/held-out split.
- Audited every locally available PLY and documented a dense-geometry validation protocol. The named VGGT/COLMAP files are sparse and the historical meshes are unlabeled, so none can support a defensible anatomical comparison; PACE was also unreachable during the check.
- Recovered real chart references for UF_Herp_3998 and UF_birds_ivory2 and compared the current linear 3 by 3 correction with a degree-2 root-polynomial model using leave-one-patch-out evaluation.
- Stress-tested the complete directory-calibration pipeline on eight real multicolour fish texture maps under two controlled camera responses. All 16 outputs completed with no skips or clipping increase.
- Updated the research database with Meeting 4, the available transcript excerpts, the latest Slack decisions, team reports, blockers, and the Week 6 handoff list.

Key results

Metric	Result	Interpretation
Held-out chart predictions	96	2 specimens × 4 camera cases
Linear mean held-out ΔE_{76}	2.573	Current PR 18 method
Root-polynomial mean ΔE_{76}	2.791	+8.4% worse
Fish textures / calibrated outputs	8 / 16	Two camera responses
Fish mean foreground ΔE_{76}	3.136 → 0.158	95.7% reduction
Worst fish-case improvement	89.2%	All cases improved
Clipping increase	0.000 pp	Both camera responses

Blockers and limits

- Absolute calibration is still waiting on the physical ColorChecker manufacturer, model, and patch-value edition. A Slack follow-up was posted September 30; no confirmed identity was available by October 2.
- Dense geometry validation still needs a labeled, densely tessellated structured-light or CT reference paired with the same specimen's photographs. The current sparse-Delaunay result is only a coarse reconstruction check.
- The fish archive contains finished OBJ/texture assets, not raw camera photographs or chart frames. This week's fish result is therefore a controlled pipeline stress test, not an independent field-calibration result.

2. Literature review

Syed asked the team to revisit neural implicit reconstruction with background masking and to identify the level of analysis expected in recent dataset papers.

Neuralangelo: high-fidelity neural surfaces

Li et al. combine multi-resolution hash grids, numerical gradients, and coarse-to-fine optimization to recover detailed surfaces from multi-view imagery. It is a strong quality-oriented candidate, but the project must verify how specimen masks and our COLMAP outputs enter its preprocessing path.

Citation: Li, Z. et al. (2023). CVPR, 8456–8465. https://openaccess.thecvf.com/content/CVPR2023/html/Li_Neuralangelo_High-Fidelity_Neural_Surface_Reconstruction_CVPR_2023_paper.html

NeuS: SDF reconstruction with volume rendering

NeuS represents a surface as the zero level set of a signed distance function and introduces a volume-rendering formulation designed to reduce geometric bias. Although the paper can reconstruct without masks, our turntable backgrounds make a masked comparison important.

Citation: Wang, P. et al. (2021). NeurIPS 34.
https://proceedings.neurips.cc/paper_files/paper/2021/hash/e41e164f7485ec4a28741a2d0ea41c74-Abstract.html

NeuS2: faster neural implicit surfaces

NeuS2 adds multi-resolution hash encoding, CUDA implementation, and progressive training, reporting approximately two orders of magnitude faster training than NeuS. That speed makes it an attractive first masked baseline if its environment can be reproduced reliably.

Citation: Wang, Y. et al. (2023). ICCV, 3295–3306. https://openaccess.thecvf.com/content/ICCV2023/html/Wang_NeuS2_Fast_Learning_of_Neural_Implicit_Surfaces_for_Multi-view_Reconstruction_ICCV_2023_paper.html

3. What I did and how I verified it

Coded-marker integration review

Ran the focused scale and mesh suites: 26 tests passed, 6 were skipped, and 3 unrelated failures came from macOS lacking `os.sched_getaffinity`. Flagged repeated short rings and one-ring captures as protocol edge cases.

Geometry-readiness audit

Classified six historical dense-looking meshes as unlabeled and the named VGGT/COLMAP outputs as sparse. Preserved the earlier 33/36 registration comparison as coarse evidence only and wrote the exact future reference-mesh protocol.

Held-out colour-model comparison

Used leave-one-patch-out predictions across two specimens and four non-reference camera cases. Root polynomial worsened mean ΔE_{76} by 0.217; the paired bootstrap 95% interval was -0.017 to $+0.454$, and three of four camera cases worsened.

Multicolour pipeline stress test

Downsampled eight real 8192×8192 RGBA fish textures to 1024 pixels, retained alpha masks, synthesized two camera responses from real fitted matrices, and invoked the full production `calibrate_directory` path. All eight preview strips were visually inspected.

Project record and coordination

Ingested the available Meeting 4 transcript excerpts and Slack sweep. Recorded Syed's request for a masked neural-implicit revisit, the need for publication-quality failure analysis, dataset-hosting decisions, and the agreed PR merge order.

Workload

Approximately 20 hours: scale review 4 h; geometry audit 3.5 h; model evaluation 5 h; fish stress test 4 h; project records 3.5 h.

4. Results visualization

Week 5 validation summary

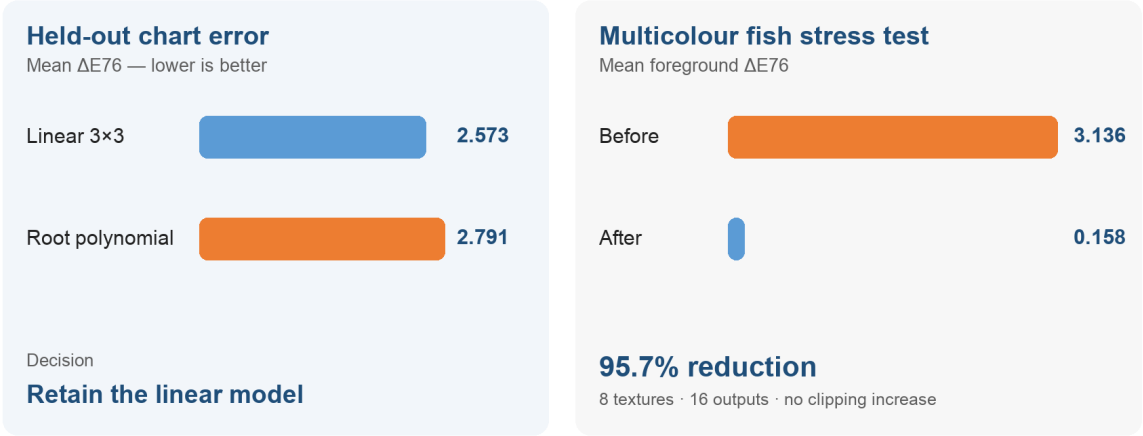


Figure 1. Held-out model selection and multicolour pipeline stress results.

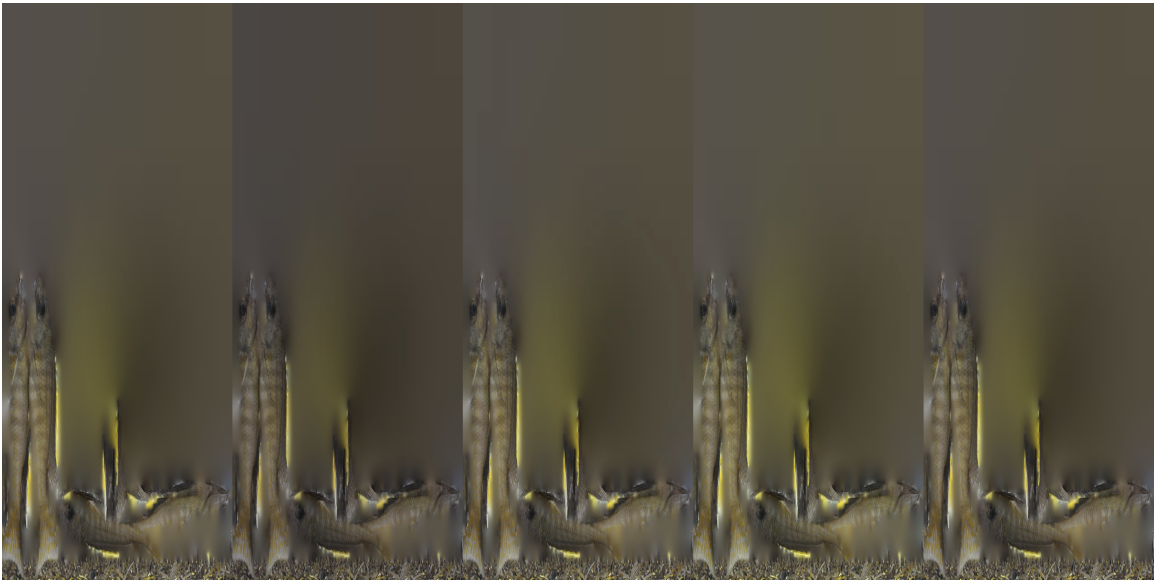


Figure 2. Representative controlled fish-texture preview strip.

5. Conclusions and decisions

- Retain the linear 3 by 3 model in PR 18. The higher-order model did not show a reliable held-out gain and was worse on average.
- Use the fish result as evidence that the implementation handles high-resolution, multicolour RGBA assets and masks; do not present it as independent biological colour validation.
- Keep colour calibration, metric scale, and geometry validation as separate provenance-controlled stages, then test their interaction after each stage is individually accepted.
- Do not make dense or anatomical geometry claims until a paired structured-light/CT reference becomes available.

Evidence and artifacts

Artifact	Location
Higher-order model report	documentation/root-polynomial-held-out-comparison-2026-10-02.md
Model comparison data	outputs/color-model-held-out-comparison-2026-10-02.json
Fish stress-test report	documentation/multicolor-fish-pipeline-stress-test-2026-10-02.md
Fish stress-test data	outputs/multicolor-fish-pipeline-stress-test-2026-10-02.json
Scale integration review	documentation/coded-marker-scale-integration-review-2026-10-02.md
Geometry readiness	documentation/color-calibration-dense-geometry-readiness-2026-10-02.md
Meeting and Slack record	documentation/meeting4-transcript.md; documentation/slack/raw/refresh-2026-10-02.md

Meeting and Slack decisions

Publication needs failure analysis, ablations, and qualitative explanations—not only a benchmark table. The dataset must be public. Slack added a masked neural-implicit revisit, and PR 18 follows PRs 19, 17, and 20 in the merge order.

Next week

- Complete the reviewer/Slack handoff for PR 18, then rebase and resolve it after PRs 19, 17, and 20 land, following Syed's requested merge order.
- Confirm ownership with Syed and Om for the neural-implicit sweep; review Neuralangelo, NeuS2, NeuS, and newer candidates, then select one reproducible masked baseline.
- Review the recent dataset/benchmark papers Syed shortlisted and extract concrete failure-analysis, ablation, and qualitative-comparison requirements for our publication.
- Compare MorphoSource and Hugging Face for public release: supported formats, 3D preview, metadata, versioning, licenses, citations, and download ergonomics.
- Rerun absolute colour calibration when the chart edition is confirmed, and run dense reference-mesh validation when a matched specimen becomes available.

Definition of a showable outcome

A reproducible masked neural-implicit baseline with command, environment, specimen output, runtime, geometry/coverage metrics, failure cases, and a concise comparison against the current photogrammetry baseline.

Open confirmations

- Ownership split with Om.
- Shared specimen and mask set.
- Primary public host and metadata/license schema.
- Chart edition and matched dense-reference availability.